to change further into the most stable Zr nucleus, in which $N - Z = 12$ ($N = 52$ and $Z = 40$), the Mo nucleus with an $N - Z$ value of 8 would first have to change, by the emission of another positron, into the Nb nucleus with an $N - Z$ value of 10—in other words, once again into a ‘doubly odd’ nucleus. This is, however, impossible from the point of view of energetics, for it would require an expenditure of energy. It is much easier for the Nb nucleus with an $N - Z$ value of 10 to change, conversely, into the Mo nucleus where $N - Z = 8$, by the emission of an electron, or into the most stable Zr nucleus where $N - Z = 12$, by the emission of a positron. Thus, a study of Figure 19 will show that in addition to the most stable nucleus, Zr, situated on the valley floor, other ‘doubly even’ nuclei with the same mass number, situated a little higher, may also be stable, whereas all the ‘doubly odd’ nuclei, situated on the upper curve, are unstable. In Figure 19, too, the possibilities of change are indicated by arrows. The arrows pointing to the lower right indicate positron emission (in certain cases, a *K*-radiation), while those pointing to the left indicate electron emission. Thus, the unstable nuclei situated on the upper curve may change by either process, if there are appropriate stable nuclei on the lower curve. An example is the potassium nucleus ${}_{19}K^{40}$, which can change by electron emission into the calcium nucleus ${}_{20}Ca^{40}$, or by positron emission into the argon nucleus ${}_{18}A^{40}$.

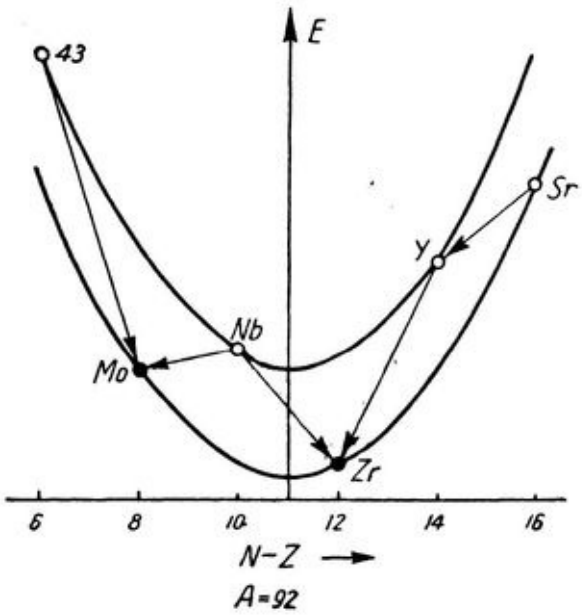


Figure 19.—Binding energy of the even nuclei.

On the ground of the evidence of these curves, we formulate therefore the following rules, to be verified by actual experiments:

1. When the mass number is an odd figure—in other words, in the case of ‘odd’ nuclei—there is only one stable nucleus for any given mass number. All